HARD · ADVANCED MATH · ANSWER KEY

Advanced Math

13

10 answers · Rationale-rich · Teach from doc

Q1 5

The correct answer is 5. The intersection points of the graphs of $y=3x^2-14x$ and $y=x$ can be found by solving the system consisting of these two equations. To solve the system, substitute x for y in the first equation. This gives $x=3x^2-14x$. Subtracting x from both sides of the equation gives $0=3x^2-15x$. Factoring $3x$ out of each term on the left-hand side of the equation gives $0=3x(x-5)$. Therefore, the possible values for x are 0 and 5. Since $y=x$, the two intersection points are $(0,0)$ and $(5,5)$. Therefore, $a=5$.

Q2 D

D. -25

Choice D is correct. Substituting b for x in the given equation yields $g\left(b+28\right)=\frac{1}{10}\left(b-a\right)^2\left(b-b\right)^2\left(b-c\right)$, which is equivalent to $g\left(b+28\right)=\frac{1}{10}\left(b-a\right)^2\left(0\right)^2\left(b-c\right)$, or $g\left(b+28\right)=0$. It follows that when $x=b+28$, $g\left(x\right)=0$. There are three points where the graph of $y=g\left(x\right)$ intersects the x-axis: $(-2,0)$, $(1,0)$, and $(3,0)$. Therefore, $g(-2)=0$, $g(1)=0$, and $g(3)=0$. It follows that either $b+28=-2$, $b+28=1$, or $b+28=3$. Subtracting 28 from each side of the equation $b+28=-2$ yields $b=-30$. Subtracting 28 from each side of the equation $b+28=1$ yields $b=-27$. Subtracting 28 from each side of the equation $b+28=3$ yields $b=-25$. Thus, the value of b must be either -30, -27, or -25. Of these values, only -25 is given as a choice.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This would be a possible value of b if the graph shown was the graph of $y=g(x+28)$, not $y=g(x)$.

Choice C is incorrect. This would be the value of c, not the value of b, if the graph shown was the graph of $y=g(x+28)$, not $y=g(x)$.

Q3

94

The correct answer is 94. Taking the square root of each side of the given equation yields $x - 47 = 1$ or $x - 47 = -1$. Adding 47 to both sides of the equation $x - 47 = 1$ yields $x = 48$. Adding 47 to both sides of the equation $x - 47 = -1$ yields $x = 46$. Therefore, the sum of the solutions to the given equation is $48 + 46$, or 94.

Q4

1728

The correct answer is 1,728. The given function can be rewritten in the form $f x = a x - h^2 + k$, where a is a positive constant and the minimum value, k, of the function occurs when the value of x is h. By completing the square, $f x = x^2 - 48x + 2{,}304$ can be written as $f x = x^2 - 48x + \frac{48}{2}^2 + 2{,}304 - \frac{48}{2}^2$, or $f x = x - 24^2 + 1{,}728$. This equation is in the form $f x = a x - h^2 + k$, where $a = 1$, $h = 24$, and $k = 1{,}728$. Therefore, the minimum value of the given function is 1,728.

Q5

C

C. 6

Choice C is correct. It's given that the graphs of the equations in the given system intersect at exactly one point, x, y, in the xy-plane. Therefore, x, y is the only solution to the given system of equations. The given system of equations can be solved by subtracting the second equation, $y = 3x + a$, from the first equation, $y = 2x^2 - 21x + 64$. This yields $y - y = 2x^2 - 21x + 64 - 3x + a$, or $0 = 2x^2 - 24x + 64 - a$. Since the given system has only one solution, this equation has only one solution. A quadratic equation in the form $rx^2 + sx + t = 0$, where r, s, and t are constants, has one solution if and only if the discriminant, $s^2 - 4rt$, is equal to zero. Substituting 2 for r, -24 for s, and $-a + 64$ for t in the expression $s^2 - 4rt$ yields $-24^2 - 4264 - a$. Setting this expression equal to zero yields $-24^2 - 4264 - a = 0$, or $8a + 64 = 0$. Subtracting 64 from both sides of this equation yields $8a = -64$. Dividing both sides of this equation by 8 yields $a = -8$. Substituting -8 for a in the equation $0 = 2x^2 - 24x + 64 - a$ yields $0 = 2x^2 - 24x + 64 + 8$, or $0 = 2x^2 - 24x + 72$. Factoring 2 from the right-hand side of this equation yields $0 = 2x^2 - 12x + 36$. Dividing both sides of this equation by 2 yields $0 = x^2 - 12x + 36$, which is equivalent to $0 = x - 6x - 6$, or $0 = x - 6^2$. Taking the square root of both sides of this equation yields $0 = x - 6$. Adding 6 to both sides of this equation yields $x = 6$.

Choice A is incorrect. This is the value of a, not x.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6

C

C. 54

Choice C is correct. It's given that the function h is defined by $hx = a^x + b$ and that the graph of $y = h(x)$ in the xy-plane passes through the points $0, 10$ and $-2, \frac{325}{36}$. Substituting 0 for x and 10 for hx in the equation $hx = a^x + b$ yields $10 = a^0 + b$, or $10 = 1 + b$. Subtracting 1 from both sides of this equation yields $9 = b$. Substituting -2 for x and $\frac{325}{36}$ for hx in the equation $h(x) = a^x + 9$ yields $\frac{325}{36} = a^{-2} + 9$. Subtracting 9 from both sides of this equation yields $\frac{1}{36} = a^{-2}$, which can be rewritten as $a^2 = 36$. Taking the square root of both sides of this equation yields $a = 6$ and $a = -6$, but because it's given that a is a positive constant, a must equal 6. Because the value of a is 6 and the value of b is 9, the value of ab is $(6)(9)$, or 54.

Choice A is incorrect and may result from finding the value of $a^{-2}b$ rather than the value of ab.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from correctly finding the value of a as 6, but multiplying it by the y-value in the first ordered pair rather than by the value of b.

Q7

A

A. 6

Choice A is correct. It's given that the graph of $y = x^2 - 9$ and line p intersect at $(1, a)$ and $(5, b)$. Therefore, the value of y when $x = 1$ is the value of a, and the value of y when $x = 5$ is the value of b. Substituting 1 for x in the given equation yields $y = (1)^2 - 9$, or $y = -8$. Similarly, substituting 5 for x in the given equation yields $y = (5)^2 - 9$, or $y = 16$. Therefore, the intersection points are $(1, -8)$ and $(5, 16)$. The slope of line p is the ratio of the change in y to the change in x between these two points: $\frac{16-(-8)}{5-1} = \frac{24}{4}$, or 6.

Choices B, C, and D are incorrect and may result from conceptual or calculation errors in determining the values of a, b, or the slope of line p.

Q8

B

B. $gx = 1444^x$

Choice B is correct. It's given that $fx = 94^x$ and $gx = fx + 2$. Substituting $x + 2$ for x in $fx = 94^x$ gives $fx + 2 = 94^{x+2}$. Rewriting this equation using properties of exponents gives $fx + 2 = 94^x 4^2$, which is equivalent to $fx + 2 = 94^x 16$. Multiplying 9 and 16 in this equation gives $fx + 2 = 1444^x$. Since $gx = fx + 2$, $gx = 1444^x$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q9 C

C. $\frac{prs}{pr + ps + rs}$

Choice C is correct. Multiplying each side of the given equation by $\frac{1}{20}$ yields $\frac{1}{20}\frac{20}{p} = \frac{1}{20}\frac{20}{q} - \frac{20}{r} - \frac{20}{s}$. Distributing $\frac{1}{20}$ on each side of this equation yields $\frac{20}{20p} = \frac{20}{20q} - \frac{20}{20r} - \frac{20}{20s}$, or $\frac{1}{p} = \frac{1}{q} - \frac{1}{r} - \frac{1}{s}$. Adding $\frac{1}{r} + \frac{1}{s}$ to each side of this equation yields $\frac{1}{s} + \frac{1}{r} + \frac{1}{p} = \frac{1}{q}$. Multiplying $\frac{1}{s}$ by $\frac{pr}{pr}$, $\frac{1}{r}$ by $\frac{ps}{ps}$, and $\frac{1}{p}$ by $\frac{rs}{rs}$ yields $\frac{pr}{prs} + \frac{ps}{prs} + \frac{rs}{prs} = \frac{1}{q}$, which is equivalent to $\frac{pr + ps + rs}{prs} = \frac{1}{q}$. Since $\frac{pr + ps + rs}{prs} = \frac{1}{q}$, and it's given that p, q, r, and s are positive, it follows that the reciprocals of each side of this equation are also equal. Thus, $\frac{prs}{pr + ps + rs} = \frac{q}{1}$, or $\frac{prs}{pr + ps + rs} = q$. Therefore, $\frac{prs}{pr + ps + rs}$ is equivalent to q.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10 C

C. $fx = -6x - 5^2 + 24$

Choice C is correct. The given equation is quadratic. The maximum value of a function defined by a quadratic equation can be displayed as a constant in the vertex form of the equation, $f(x) = a(x - h)^2 + k$, where the maximum value of the function is k, which occurs when $x = h$, and a is a constant. The given equation can be rewritten in this form by completing the square. To complete the square, the given equation can be rewritten as $f(x) = -6(x^2 - 10x) - 126$, which is equivalent to $f(x) = -6(x^2 - 10x + 25) - 126 + 6(25)$, or $f(x) = -6(x - 5)^2 + 24$. This equation is in vertex form, where $a = -6$, $h = 5$, and $k = 24$. Therefore, $f(x) = -6(x - 5)^2 + 24$ displays the maximum value, 24, of the function as a constant.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This form displays one of the zeros, not the maximum value, of the function as a constant.

Choice D is incorrect. This form displays the zeros, not the maximum value, of the function as constants.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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